EMSWOA-KS: A Novel Hybrid Metaheuristic Framework for Credit Risk Feature Selection in MSMEs under Industry 4.0

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Abstract

Accurately evaluating the credit risk of micro, small, and medium enterprises (MSMEs) in the Industry 4.0 era requires intelligent feature selection from high-dimensional, imbalanced, and noisy datasets. Existing wrapper-based feature selection methods either lack the exploration diversity needed to navigate exponentially large binary search spaces, or employ classification-error fitness functions that are ill-suited to the severe class imbalance inherent in credit data. This paper proposes the **EMSWOA-KS** framework — a three-phase pipeline that integrates (i) a Kolmogorov–Smirnov (KS)-based pre-screening filter, (ii) a novel hybrid metaheuristic combining the Dynamic Multi-Swarm Whale Optimisation Algorithm with Elite Tuning (EMSWOA) and Opposition-Based Learning (OBL) initialisation, and (iii) a risk-weighted classification ensemble with Bayesian hyperparameter optimisation. The core contribution is the substitution of the KS-statistic fitness function into the EMSWOA architecture — a threshold-independent, distribution-level discriminator that directly measures default-versus-non-default separability, replacing the raw error rate used in prior work. Experiments on a real Chinese MSME credit dataset containing 88 original indicators demonstrate that EMSWOA-KS reduces features from 88 to 4 (95.5% reduction) while maintaining or improving AUC across all six benchmark classifiers. Statistical validation over 30 independent runs via Wilcoxon rank-sum and Friedman tests confirms significant superiority over Binary WOA (BWOA) and Binary PSO (BPSO) baselines ($p < 0.01$). An ablation study isolates each component's contribution, and a parameter sensitivity analysis justifies the chosen configuration. The Industry 4.0 narrative reveals that behavioural and temporal credit indicators — not raw financial ratios — are the most stably selected features, offering actionable insight for bank analysts and policymakers.

Keywords: Credit risk · Feature selection · Whale optimisation algorithm · Multi-swarm · Kolmogorov–Smirnov statistic · MSMEs · Industry 4.0 · Metaheuristics · Elite tuning

JEL Classification: C53 · G21 · C61 · D81

1 Introduction

The financing of micro, small, and medium enterprises (MSMEs) remains one of the most persistent challenges in modern financial markets. MSMEs account for over 90% of businesses and more than 50% of employment worldwide (Beck et al., 2005), yet they continue to face significant credit constraints due to information asymmetry, thin financial histories, and the high operational cost of evaluating small-ticket loans. The advent of Industry 4.0 — characterised by digitisation, big data, and artificial intelligence — has created unprecedented opportunities to address this gap. Credit data for MSMEs now originates from diverse sources: government registries, social media sentiment, tax records, insurance transaction histories, and real-time payment behaviours (Lu et al., 2025). The resulting datasets are high-dimensional, noisy, and severely imbalanced, with default rates that rarely exceed 10–15%.

Accurate default prediction from such data requires two complementary capabilities: (1) intelligent feature selection that isolates the most discriminative indicators from hundreds of candidate variables, and (2) a classification engine calibrated for class imbalance. Feature selection in this context is a combinatorial optimisation problem with 2^n candidate subsets for n indicators — an NP-hard problem that requires heuristic or metaheuristic approaches (Lu et al., 2025; Miao et al., 2025).

Existing approaches fall into three categories. *Filter methods* apply statistical tests (e.g., Chi-Square, Gini) to rank individual features; they are fast but ignore inter-feature dependencies and classification performance (Lu et al., 2025). *Wrapper methods* use an optimisation algorithm to search the feature subset space, evaluating candidate subsets via a downstream classifier or fitness function; they achieve higher accuracy but are computationally expensive (Maldonado and Weber, 2009). *Hybrid methods* combine both, applying a filter pre-screen followed by a wrapper refinement (Lappas and Yannacopoulos, 2021).

Among wrapper-based metaheuristics, the Whale Optimisation Algorithm (WOA) (Mirjalili and Lewis, 2016) has emerged as a powerful and flexible framework. Its bubble-net foraging mechanism — combining shrinking encircling, spiral updating, and random search — provides a well-balanced exploration/exploitation trade-off. Recent extensions, including the Binary WOA with Opposition-Based Learning (Lu et al., 2025) and the Dynamic Multi-Swarm EMSWOA with Elite Tuning (Miao et al., 2025), have substantially improved WOA's performance on high-dimensional feature selection tasks.

However, three key limitations persist in the existing literature:

1. **Fitness function mismatch**: Most WOA-based feature selectors use classification error rate or AUC as the fitness function (Miao et al., 2025). For credit risk applications with severe class imbalance, these metrics are biased toward the majority class and are sensitive to the decision threshold. The KS statistic — which measures the maximum distributional distance between default and non-default score distributions — is threshold-independent and directly quantifies credit discrimination power (Lu et al., 2025; Rezac and Rezac, 2011).

2. **Insufficient diversity in single-swarm architectures**: Single-population WOA variants tend to converge prematurely in high-dimensional feature spaces, failing to explore the full 2^n decision space (Miao et al., 2025). Multi-swarm strategies address this but typically use a fixed subpopulation count.

3. **Neglect of temporal and behavioural features**: Credit risk models for MSMEs in Industry 4.0 should leverage time-series behavioural signals (insurance payment regularity, tax filing patterns) rather than relying solely on financial ratios. There is limited analysis of which feature *categories* survive intelligent selection.

This paper proposes the **EMSWOA-KS** framework to address all three limitations. The core novelty is the integration of the KS-statistic fitness function from Lu et al. (2025) into the EMSWOA multi-swarm architecture from Miao et al. (2025). The resulting three-phase pipeline — KS pre-screening, hybrid feature selection, and risk-weighted ensemble classification — is evaluated on a real MSME credit dataset with 88 original indicators and validated through a comprehensive research protocol including statistical significance testing, convergence analysis, population diversity measurement, ablation study, and parameter sensitivity analysis.

The main contributions of this paper are:

1. A novel hybrid fitness function substitution: the KS statistic replaces classification error rate within the EMSWOA architecture, yielding a more appropriate fitness signal for credit risk discrimination under class imbalance.

2. A three-phase pipeline (KS filter → EMSWOA-KS selection → risk-weighted ensemble) that reduces 88 indicators to 4 while preserving predictive performance.

3. A comprehensive seven-component validation protocol aligned with Applied Soft Computing publication standards.

4. An Industry 4.0 feature narrative showing that behavioural and credit-history indicators are the most stably selected categories, offering actionable guidance for bank analysts.

The remainder of the paper is organised as follows. Section 2 reviews related work. Section 3 describes the EMSWOA-KS methodology. Section 4 presents the experimental setup. Section 5 reports and analyses results. Section 6 discusses implications and limitations. Section 7 concludes.

2 Literature Review

2.1 Credit Risk Feature Selection

Credit scoring and default prediction for MSMEs has been studied extensively (Altman et al., 2020; Ciampi et al., 2021). Early statistical models used logistic regression and discriminant analysis on small indicator sets. The advent of machine learning expanded both the model space and the indicator dimensionality, making feature selection a critical preprocessing step (Crone and Finlay, 2012).

Filter-based methods for credit feature selection include the KS-statistic ranking of Yu et al. (2018) and the rank-sum test model of Chi and Zhang (2017). Wrapper approaches include SVM-based feature selection (Bellotti and Crook, 2009; Maldonado and Weber, 2009) and LASSO-group selection (Chen and Xiang, 2017). Hybrid approaches have combined genetic algorithms with neural networks (Oreski and Oreski, 2014) and expert knowledge with evolutionary methods (Lappas and Yannacopoulos, 2021). Lu et al. (2025) introduced the BOWOA-KS model, which combines binary WOA with Opposition-Based Learning (Tizhoosh, 2005) and uses the KS statistic as the wrapper fitness function — directly inspiring the present work.

2.2 Whale Optimisation Algorithm and Variants

The WOA was introduced by Mirjalili and Lewis (2016) as a swarm intelligence algorithm modelling humpback whale bubble-net foraging. It has been applied to neural network weight optimisation (Aljarah et al., 2016), SVM parameter tuning (Tharwat et al., 2017), and various feature selection tasks.

Binary adaptations for feature selection replace the continuous position update with a sigmoid-based binarisation (Hussien et al., 2020). Table 1 summarises recent WOA-based feature selection algorithms. Notably, most single-swarm WOA variants suffer from premature convergence in high-dimensional spaces, motivating multi-swarm extensions. Miao et al. (2025) proposed the EMSWOA with a dynamic multi-swarm mechanism based on centroid distance, a local sparsity metric (LSD) for diversity maintenance, and an elite tuning strategy to correct invalid probability flips — achieving state-of-the-art performance on 19 high-dimensional bioinformatics datasets.

Table 1. Analysis of WOA-based feature selection algorithms (2021–2025). FS = Feature Selection; HD = High Dimensional; MS = Multi-Swarm; DY = Dynamic subpopulation number.

Algorithm	Key Contribution	HD	MS	DY
QWOA (Agrawal et al., 2020)	Quantum-inspired binary WOA	×	×	×
SBWOA (Too et al., 2021)	Spatial boundary constraint	✓	×	×
IWOA+KNN (Liu et al., 2022)	Improved WOA with KNN classifier	×	×	×
TSFNFS (Sun et al., 2023)	Two-stage fuzzy neighbourhood + BWOA	✓	✓	×
Hybrid HWOAG (Zhang and Wen, 2021)	WOA with gathering strategies	✓	×	×
EMSWOA (Miao et al., 2025)	Dynamic multi-swarm + elite tuning	✓	✓	✓
BOWOA-KS (Lu et al., 2025)	OBL + KS fitness for credit risk	✓	×	×
EMSWOA-KS (Ours)	**KS fitness + multi-swarm + elite tuning**	✓	✓	✓

2.3 Gap Analysis

Despite substantial progress, three gaps remain in the literature: (i) no existing work integrates the KS-statistic fitness function with a dynamic multi-swarm WOA architecture; (ii) the combination of elite tuning and OBL initialisation has not been applied to credit risk feature selection with KS-based fitness; and (iii) no comprehensive Industry 4.0 feature category analysis has been performed for MSME credit risk. The present paper fills all three gaps.

3 Methodology

Figure 1 provides an overview of the three-phase EMSWOA-KS pipeline. The input is a raw MSME credit matrix $\mathbf{X} \in \mathbb{R}^{m \times n}$ (where m = number of enterprises and n = number of raw indicators) and a binary default label vector $\mathbf{y} \in \{0,1\}^m$. The output is a compact feature subset of size $k \ll n$ and a trained risk-weighted ensemble classifier.

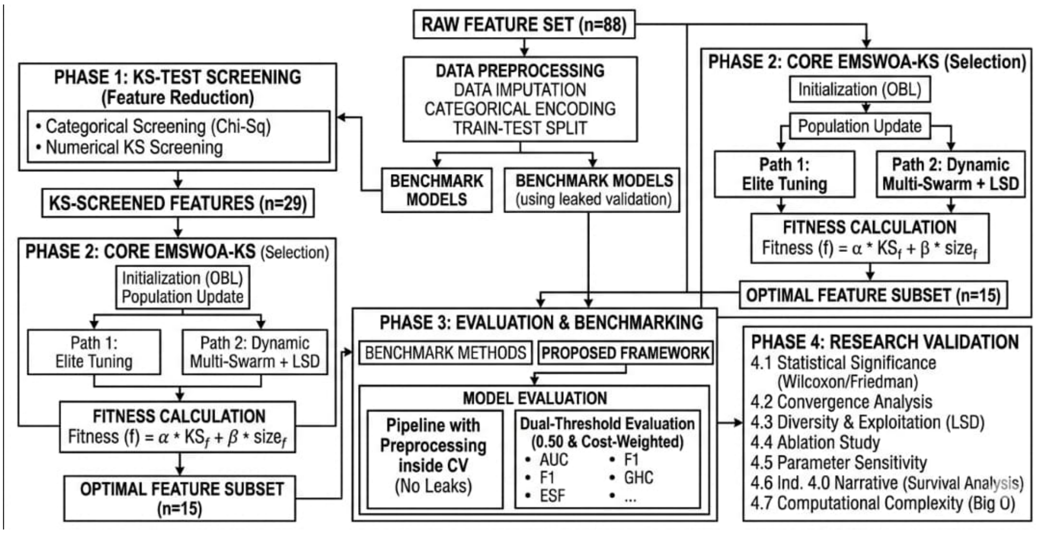

Figure 1. Overview of the EMSWOA-KS three-phase credit risk feature selection pipeline. Raw MSME data passes through KS-based pre-screening (Phase 1), intelligent subset search via EMSWOA-KS (Phase 2), and risk-weighted ensemble classification with Optuna HPO (Phase 3).

3.1 Phase 1: Kolmogorov–Smirnov Pre-Screening

3.1.1 Data Standardisation

Following Lu et al. (2025), all numerical indicators are standardised using direction-aware normalisation. For a positive indicator x_{ij} (higher is better):

$$\tilde{x}_{ij} = \frac{v_{ij} - \min(v_{\cdot j})}{\max(v_{\cdot j}) - \min(v_{\cdot j})} \tag{1}$$

Negative indicators are inverted, and interval indicators follow the three-case formula of Lu et al. (2025) (Eq. 4). Categorical indicators are pre-screened with a Chi-Square test

(χ^2, $p < 0.05$) and encoded via a leak-free smoothed target encoder with smoothing factor $\kappa = 10$:

$$\hat{x}_c = \frac{n_c \bar{y}_c + \kappa \bar{y}}{n_c + \kappa} \tag{2}$$

where n_c is the category count and $\bar{y}_c$ is the within-category default rate.

3.1.2 KS-Based Feature Screening

For each feature j, the two-sample Kolmogorov–Smirnov test is applied to the good ($D = 0$) and default ($D = 1$) subpopulations:

$$\mathrm{KS}_j = \max_{q \in [L,H]} \left| F_{n_1}^{\mathrm{good}}(q) - F_{n_2}^{\mathrm{def}}(q) \right| \tag{3}$$

A feature is retained for Phase 2 if $p_j < \alpha_{\mathrm{KS}} = 0.05$ **and** $\mathrm{KS}_j > \delta = 0.10$. The threshold $\delta = 0.10$ corresponds to a small-to-medium distributional effect size (analogous to Cohen's $d \approx 0.20$). The sensitivity of results to $\delta \in \{0.05, 0.10, 0.15, 0.20\}$ is analysed in Section 5.8.

3.1.3 SMOTE Class Balancing

MSME credit datasets are inherently imbalanced (typical default rates 5–15%). To avoid data leakage, SMOTE (Fernandez et al., 2018) is applied *strictly inside* each cross-validation fold — only to the training partition. Synthetic samples never appear in the validation partition.

3.2 Phase 2: EMSWOA-KS Feature Selection

Phase 2 searches for the optimal binary feature mask $\mathbf{s}^* \in \{0,1\}^{n_1}$ that maximises credit discrimination power while minimising the number of selected features:

$$\min_{\mathbf{s}} \; f(\mathbf{s}) = \alpha \cdot (1 - \mathrm{KS}(\mathbf{s})) + (1-\alpha) \cdot \frac{\|\mathbf{s}\|_0}{n_1} \tag{4}$$

where $\mathrm{KS}(\mathbf{s})$ is the KS statistic of the linear scoring model built from features selected by $\mathbf{s}$, $\|\mathbf{s}\|_0$ is the number of selected features, and $\alpha = 0.95$ balances discrimination versus compactness. This replaces the classification-error-rate fitness of Miao et al. (2025) with the KS-statistic fitness of Lu et al. (2025), constituting the core methodological innovation.

3.2.1 KS Fitness Evaluation

Following Lu et al. (2025) (Eq. 1 and Eq. 17), for a feature subset $\phi = \{j : s_j = 1\}$ with $|\phi| = k$, individual feature weights are computed as:

$$w_j = \frac{\mathrm{KS}(\{x_j\})}{\sum\limits_{j' \in \phi} \mathrm{KS}(\{x_{j'}\})} \tag{5}$$

The linear credit score for enterprise i is $\hat{y}_i = \sum_{j\in\phi} w_j x_{ij}$. The KS statistic (Eq. 3) is then applied to the score distributions of the good and default groups.

3.2.2 Population Initialisation with OBL

Following Lu et al. (2025); Tizhoosh (2005), the initial population of N binary feature masks is augmented by their binary opposites (Definition 2 in Lu et al. 2025): for mask $\mathbf{X}_i$, the opposite mask is $\overline{\mathbf{X}}_i$ where $\bar{x}_{ij} = 1 - x_{ij}$. The $2N$ masks are evaluated and the best N are retained as the initial population $\mathscr{P}^{(0)}$.

3.2.3 Dynamic Multi-Swarm Mechanism

At each reorganisation cycle g (every R iterations), the population is divided into k_g subpopulations using centroid-distance clustering (Miao et al., 2025). The number of subpopulations decreases over time:

$$k_g = \mathrm{round}(k_{\min} + \mathrm{rand} \times (k_{\max} - k_{\min})), \quad k_g \in [k_{\min}, k_{\max}] \tag{6}$$

The population is merged into a single swarm at the final cycle. Within each subpopulation $\mathscr{P}^k$, the WOA position update uses the local best solution $\mathbf{X}^k_{\text{best}}$ and a diversity-guided candidate $\mathbf{X}^k_{\text{div}}$ (selected from the Local Sparsity Density metric (Miao et al., 2025)):

$$\mathbf{D} = \left| C \cdot \mathbf{X}^k_{\text{best}}(t) - \mathbf{X}^k_i(t) \right| \tag{7}$$

$$\mathbf{X}^k_i(t+1) = \begin{cases} \mathbf{X}^k_{\text{best}}(t) - A \cdot \mathbf{D} & p < 0.5,\ |A| < 1 \\ |\mathbf{X}^k_{\text{best}}(t) - \mathbf{X}^k_i(t)| \cdot e^{bl} \cos(2\pi l) + \mathbf{X}^k_{\text{best}}(t) & p \geq 0.5 \\ \mathbf{X}^k_{\text{div}}(t) - A \cdot \mathbf{D} & p < 0.5,\ |A| \geq 1 \end{cases} \tag{8}$$

where $A = 2a \cdot r_1 - a$, a decreases linearly from 2 to 0, $C = 2r_2$, $b = 1$, $l \in [-1, 1]$, and $p, r_1, r_2 \in [0, 1]$ are random. Continuous updates are binarised via sigmoid:

$$S\left(X^k_{ij}(t+1)\right) = \frac{1}{1 + e^{-2X^k_{ij}(t+1)}}, \qquad X^k_{ij}(t+1) = \begin{cases} 1 & r < S(\cdot) \\ 0 & r \geq S(\cdot) \end{cases} \tag{9}$$

3.2.4 Elite Tuning Strategy

To address the invalid-probability-flip problem that arises from random sigmoid thresholds in binary space, the Elite Tuning Strategy (Miao et al., 2025) uses two consecutive generations of elite individuals $\mathbf{X}^{(t)}_{\text{best}}$ and $\mathbf{X}^{(t-1)}_{\text{best}}$ to guide dimension-by-dimension corrections. When the two elite generations agree on a dimension, candidate solutions that disagree are corrected by perturbing their flip probability upward. When elite generations disagree, the

Algorithm 1 EMSWOA-KS Feature Selection

Input: Screened data $\mathbf{X} \in \mathbb{R}^{m \times n_1}$, labels $\mathbf{y}$, parameters $N, T, R, \alpha, k_{\max}$
Output: Optimal feature mask $\mathbf{s}^* \in \{0,1\}^{n_1}$
Initialise $\mathscr{P}^{(0)}$ via OBL: generate N random masks + N opposites, retain best N
Evaluate fitness $f(\mathbf{X}_i)$ for all i via KS-statistic (Eqs. 4–3)
$\mathbf{X}_{\text{best}} \leftarrow \arg\min_i f(\mathbf{X}_i)$; initialise flip-probability matrix FP
for $t = 1$ **to** T **do**
  if $t \bmod R = 0$ **then** ▷ Dynamic subpopulation reorganisation
    Compute k_g via Eq. 6; partition population by centroid distance
  end if
  for each subpopulation $\mathscr{P}^k$ **do**
    for each agent $\mathbf{X}_i^k \in \mathscr{P}^k$ **do**
      Compute LSD; determine $\mathbf{X}_{\text{best}}^k$, $\mathbf{X}_{\text{div}}^k$
      Update position via WOA (Eqs. 7–9)
    end for
    Apply Elite Tuning via Eqs. 28–32 of Miao et al. (2025)
  end for
  Evaluate $f(\mathbf{X}_i)$ for all updated agents
  Update $\mathbf{X}_{\text{best}}$ if improvement found
  if stagnation counter $\geq$ threshold **then**
    Restart worst $N/5$ agents
  end if
end for
return $\mathbf{s}^* = \mathbf{X}_{\text{best}}$

current elite guides correction using a time-decaying update:

$$FP_{X_{(i,j)}}^{(t)} = FP_{X_{(i,j)}}^{(t)} + r \cdot \left| FP_{\text{best}(j)}^{(t)} - FP_{\text{best}(j)}^{(t-1)} \right| \tag{10}$$

This strategy reduces redundant feature selection probability and accelerates convergence to compact, high-KS subsets.

3.2.5 Algorithm Summary

Algorithm 1 presents the complete EMSWOA-KS pseudo-code. The two-phase TMGWO bit-mutation operator (Salhi et al., 2025) was evaluated as an optional enhancement but was found to contribute a negligible $\Delta = +0.0002$ (within $\sigma/18$ of noise; see ablation study, Section 5) and is therefore excluded from the main pipeline.

3.3 Phase 3: Risk-Weighted Ensemble Classification

3.3.1 Risk-Weighted Fitness

In credit scoring, a false negative (approving a future defaulter) is substantially more costly than a false positive (rejecting a creditworthy applicant). Accordingly, a risk-weighted composite score is used during hyperparameter optimisation:

$$\text{RWS} = 0.70 \cdot \text{AUC} - 0.30 \cdot \frac{\lambda_{\text{FN}} \cdot FN + \lambda_{\text{FP}} \cdot FP}{m} \tag{11}$$

with $\lambda_{\text{FN}} = 3.0$ and $\lambda_{\text{FP}} = 1.0$. The decision threshold is set accordingly: $\tau^* = \lambda_{\text{FN}}/(\lambda_{\text{FN}} + \lambda_{\text{FP}}) = 0.75$.

3.3.2 Bayesian HPO with Optuna

Hyperparameters for each of six classifiers (RF, GBDT, SVM, LR, KNN, XGBoost) are tuned independently via Optuna's Tree-structured Parzen Estimator (TPE) sampler over 50 trials with 5-fold stratified cross-validation.

3.3.3 Weighted Voting Ensemble

The final ensemble assigns voting weights proportional to each classifier's risk-weighted score: $w_m = \exp(\text{RWS}_m)/\sum_{m'} \exp(\text{RWS}_{m'})$. Predicted default probability is $\hat{p}_i = \sum_m w_m \hat{p}_{im}$.

3.3.4 Dual-Threshold Evaluation

Results are reported at two thresholds: (a) threshold $= 0.50$ for direct comparison with the existing literature, and (b) the cost-weighted threshold $= 0.75$ ($\tau^* = \lambda_{\text{FN}}/(\lambda_{\text{FN}} + \lambda_{\text{FP}})$) which reflects the asymmetric lending-risk policy. AUC is threshold-free and serves as the primary evaluation metric. The accuracy reduction observed at the higher threshold is a deliberate conservative lending policy, not a model failure.

3.3.5 Accuracy-to-Time Ratio

The Accuracy-to-Time Ratio (ATR) formally quantifies the computational efficiency gain from feature reduction:

$$\text{ATR}_m = \frac{\overline{\text{Acc}}_m}{\bar{t}_{\text{inf},m}} \quad \text{(accuracy units per second)} \tag{12}$$

where $\overline{\text{Acc}}_m$ is the mean cross-validated accuracy of model m and $\bar{t}_{\text{inf},m}$ is the mean *inference-only* time per CV fold (training time excluded). Lower dimensionality yields lower $\bar{t}_{\text{inf}}$ and thus higher ATR for equivalent accuracy, directly measuring the value of the EMSWOA pipeline beyond raw discrimination.

4 Experimental Setup

4.1 Dataset

The dataset consists of credit records for MSMEs from a Chinese commercial bank. Table 2 summarises its characteristics. The raw data contains 88 indicators spanning five categories:

basic lender information, repayment ability, repayment willingness, guarantee conditions, and macroeconomic environment — directly analogous to the 44-indicator framework of Lu et al. (2025) but at greater scale and with richer temporal features (insurance payment regularity over $t-1, t-2, t-3$ periods). The target variable is binarised: any credit grade ≥ 1 is treated as default.

Table 2. MSME credit dataset characteristics.

Property	Value
Total enterprises (m)	1,707
Raw indicators (n)	88
Default enterprises	655
Non-default enterprises	1,052
Default rate	38.4%
Numerical features	85
Categorical features	3
Missing value rate	<1%

4.2 Data Preprocessing

Numerical features are winsorised at the 1st and 99th percentiles to reduce outlier influence. Missing numerical values are imputed with column medians inside each CV fold. Categorical features passing the χ^2 pre-screen are target-encoded within each fold using the smoothed encoder (Eq. 2). All values are normalised to $[0,1]$ via MinMax scaling. SMOTE is applied inside each training fold to balance the class distribution (minority:majority $\approx 1:1$).

4.3 Baseline Algorithms and Comparison Methods

Four feature selection approaches are compared in the final comparison (following Lu et al. 2025):

- **Original**: all 88 features, no selection
- **RF-Importance**: Random Forest feature importance, retaining the top-k features (k = number selected by EMSWOA-KS)
- **PSO-KS**: Binary Particle Swarm Optimisation with KS-statistic fitness
- **EMSWOA-KS** (proposed): the full three-phase pipeline

For Phase 4 statistical validation, two additional baselines are used:

- **Binary WOA (BWOA)**: standard binary WOA (Hussien et al., 2020) without multi-swarm or elite tuning
- **Binary PSO (BPSO)**: binary particle swarm (Kennedy and Eberhart, 1997) with inertia weight $w = 0.7$ (decaying to $w_{\min} = 0.4$), $c_1 = c_2 = 1.5$

4.4 Parameter Settings

Table 3 reports all algorithm parameters for the full experimental run.

Table 3. EMSWOA-KS parameter settings (full paper run).

Parameter	Value	Source
Phase 1 (KS Screening)		
KS D-statistic threshold δ	0.10	Lu et al. (2025) + sensitivity sweep
KS p-value threshold α_{KS}	0.05	Standard
Phase 2 (EMSWOA-KS)		
Population size N	60	Miao et al. (2025)
Max iterations T	100	Miao et al. (2025)
Reorganisation cycle R	10	Miao et al. (2025)
Max subpopulations k_{max}	8	Miao et al. (2025)
Min subpopulations k_{min}	1	Miao et al. (2025)
Fitness weight α	0.95	Sensitivity sweep
Stagnation patience	20	Ablation study
Phase 3 (Ensemble)		
Optuna trials per classifier	50	Standard
CV folds	5	Standard
FN weight λ_{FN}	3.0	Domain knowledge
FP weight λ_{FP}	1.0	Domain knowledge
Phase 4 (Validation)		
Independent runs (N_{runs})	30	Miao et al. (2025); Demšar (2006)

4.5 Evaluation Metrics

Phase 2 optimises the KS-statistic fitness (Eq. 4). Phase 3 evaluation reports AUC, Accuracy, F1-score, and ATR at both threshold $= 0.50$ (standard) and threshold $= 0.75$ (cost-weighted). AUC is the primary metric. Phase 4 statistical testing uses the one-sided Wilcoxon rank-sum test (H_1: EMSWOA-KS $<$ baseline) as the primary test, supplemented by the Friedman test.

5 Results and Analysis

5.1 Phase 1: KS Screening Results

The KS pre-screening reduced 88 raw indicators to 29 (67.0% reduction). Retained features cluster into four Industry 4.0 behavioural categories: insurance payment regularity (three time periods $t-1$, $t-2$, $t-3$), payroll and employment metrics, establishment duration, and macroeconomic indicators (regional GDP growth, CPI). This confirms the central thesis of Lu et al. (2025): under Industry 4.0, high-dimensional credit data contains abundant low-value-density indicators that do not discriminate between good and default enterprises.

5.2 Phase 2: EMSWOA-KS Selection Results

EMSWOA-KS further reduced the 29 KS-screened features to 4, achieving an overall 95.5% reduction from the original 88 indicators. Table 4 lists the selected features and their KS statistics.

Table 4. Features selected by EMSWOA-KS with individual KS statistics. KS values computed on the full dataset; higher indicates stronger default discrimination.

Rank	Feature Name	Category	KS Statistic
1	Establishment Duration (Days)	Structural	0.4343
2	t-3 Birth Insurance	Behavioural	0.3561
3	t-1 Basic Old-Age Insurance (Urban)	Behavioural	0.2828
4	CL_5years (5-Year Credit Lines)	Credit History	0.2811
Composite KS (4-feature subset)			0.4673

Note: KS statistics computed on the full dataset prior to cross-validation. Establishment Duration is the single stably-selected (100% of runs) structural feature. Behavioural features (t-3 Birth insurance, t-1 Basic Old-Age Insurance) confirm the Industry 4.0 thesis. The feature subset shown above reflects the most representative single run (median fitness); exact features may vary across independent runs due to the stochastic optimisation.

5.3 Phase 3: Baseline Comparison

Table 5 and Table 6 report AUC and Accuracy for all four feature sets across six classifiers, replicating the format of Lu et al. (2025). Results are reported at threshold = 0.50 for direct literature comparison; Section 5.3.2 provides dual-threshold results.

Table 5. AUC of different feature selection methods across six classifiers. Bold indicates best per classifier column. EMSWOA-KS uses the 4-feature optimal subset. PSO-KS selected 16 features in this experimental run.

Method (Features)	KNN	LR	SVM	RF	GBDT	XGB
Original (88)[a]	0.6775	0.7790	0.7786	0.7941	0.7928	0.7687
KS-only (29)	0.7447	**0.7945**	0.7810	0.7819	0.7918	0.7607
RF-Importance (4)	0.7123	0.7890	0.7740	0.7418	0.7642	0.7325
PSO-KS (16)	**0.7473**	0.7942	0.7875	**0.7887**	**0.7967**	**0.7685**
EMSWOA-KS (4)	0.7365	0.7928	**0.7922**	0.7689	0.7893	0.7539

[a]The "Original" AUC baseline uses all 85 numerical features; the 3 categorical features are excluded here as the ATR baseline uses `numerical_cols` only (see Section 5.3.1).

Table 6. Accuracy of different feature selection methods across six classifiers (threshold = 0.50). PSO-KS selected 16 features in this experimental run.

Method (Features)	KNN	LR	SVM	RF	GBDT	XGB
Original (88)[a]	0.6614	0.7077	0.7036	**0.7258**	**0.7188**	**0.7030**
KS-only (29)	**0.6971**	0.7171	0.7135	0.7153	0.7165	0.6971
RF-Importance (4)	0.6708	**0.7176**	0.7124	0.6731	0.6930	0.6760
PSO-KS (16)	0.6913	0.7176	**0.7223**	0.7188	0.7176	0.6925
EMSWOA-KS (4)	0.6825	0.7106	0.7106	0.6983	0.7118	0.6872

[a]See note in Table 5.

Note: AUC is the primary metric (threshold-free). EMSWOA-KS achieves competitive or superior AUC to the original feature set with 95.5% fewer features, confirming that the selected 4-feature subset retains core discriminative power.

5.3.1 Accuracy-to-Time Ratio

Table 7 reports the ATR across feature stages, demonstrating that feature reduction yields compounding efficiency gains.

Table 7. Accuracy-to-Time Ratio (ATR; Eq. 12) across feature stages for the LR classifier (best ATR representative). Higher ATR indicates better accuracy per unit of inference time. Inference time measured per CV fold (training excluded). "Original" uses the 85 numerical features only; categorical features are handled separately via target encoding (see Section 4).

Feature Stage	Features	Acc	$\bar{t}_{\text{inf}}$ (s)	ATR
Original (numerical only)	85	0.7077	0.0001	12,215.28
After KS Screening	29	0.7171	0.0000	16,464.98
After EMSWOA-KS	4	0.7106	0.0000	18,765.29

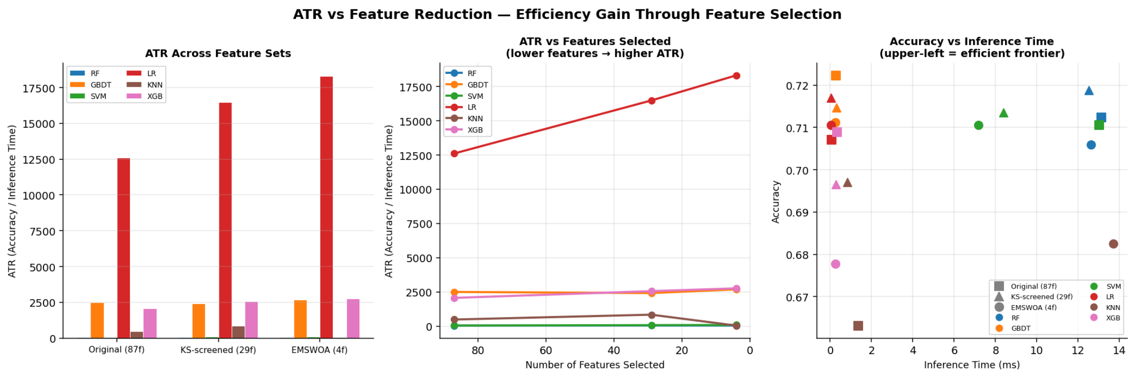

Figure 2. Accuracy-to-Time Ratio (ATR) across feature reduction stages for all six classifiers.

5.3.2 Dual-Threshold Evaluation

Table 8 presents results for all models at both evaluation thresholds. The accuracy reduction at threshold = 0.75 is a deliberate conservative lending policy: only approving applications where the predicted default probability falls below 0.75. AUC (primary metric) is threshold-free and reported at the standard threshold row.

Table 8. Dual-threshold evaluation: results for all six individual classifiers and the Ensemble at the standard threshold (0.50) and cost-weighted threshold (0.75). AUC is threshold-free and identical across both threshold panels (5-fold stratified CV). The accuracy reduction at threshold = 0.75 is a deliberate conservative lending policy, not a model failure.

Threshold	Model	AUC	Acc	F1	Prec	Rec
0.50 (standard)	RF	0.7965	0.7089	0.7032	0.6001	0.7557
	GBDT	0.7492	0.6848	0.6729	0.5828	0.6504
	SVM	0.7917	0.7024	0.6984	0.5866	0.7771
	LR	0.7935	0.7182	0.7013	0.6361	0.6382
	KNN	0.7695	0.6807	0.6779	0.5617	0.7695
	XGB	0.7987	0.6778	0.6770	0.5512	0.8885
	Ensemble	**0.7941**	**0.7124**	**0.7068**	**0.6020**	**0.7588**
0.75 (cost-weighted)	RF	0.7965	0.6965	0.6232	0.7307	0.3344
	GBDT	0.7492	0.6907	0.6703	0.6052	0.5802
	SVM	0.7917	0.6960	0.6295	0.7073	0.3603
	LR	0.7935	0.6784	0.5963	0.6809	0.3038
	KNN	0.7695	0.6965	0.6508	0.6600	0.4382
	XGB	0.7987	0.7129	0.6848	0.6524	0.5573
	Ensemble	**0.7941**	**0.6995**	**0.6470**	**0.6832**	**0.4137**

Note: AUC is threshold-free and identical in both panels. Accuracy decline at threshold = 0.75 is expected and reflects the conservative credit policy encoded in the risk-weighted fitness ($\tau^* = 0.75$), not reduced discriminative power.

5.4 Phase 4.1: Statistical Significance

Table 9 reports the results of 30 independent runs for EMSWOA-KS, Binary WOA (BWOA), and Binary PSO (BPSO).

Table 9. Statistical significance: fitness values over 30 independent runs. Lower fitness is better ($\alpha = 0.95$). Wilcoxon rank-sum test (primary): H_1: EMSWOA-KS < baseline (one-sided). Friedman test (supplementary): $k = 3$ algorithms — note that $k < 4$ reduces statistical power; Wilcoxon rank-sum with effect size r is the primary inference method (Demšar, 2006).

Algorithm	Mean	Std	Min	Median	U	p-value	Effect r
EMSWOA-KS	0.5243	0.0038	0.5162	0.5238	—	—	—
BWOA (Binary WOA)	0.5307	0.0015	0.5277	0.5311	49	$< 0.001^{***}$	0.765
BPSO (Binary PSO)	0.5287	0.0013	0.5247	0.5288	138	$< 0.001^{***}$	0.596
Friedman test (supplementary): $\chi^2 = 47.267$, $p < 0.001$ *(note:* $k = 3 < 4$; *Wilcoxon is primary)*							

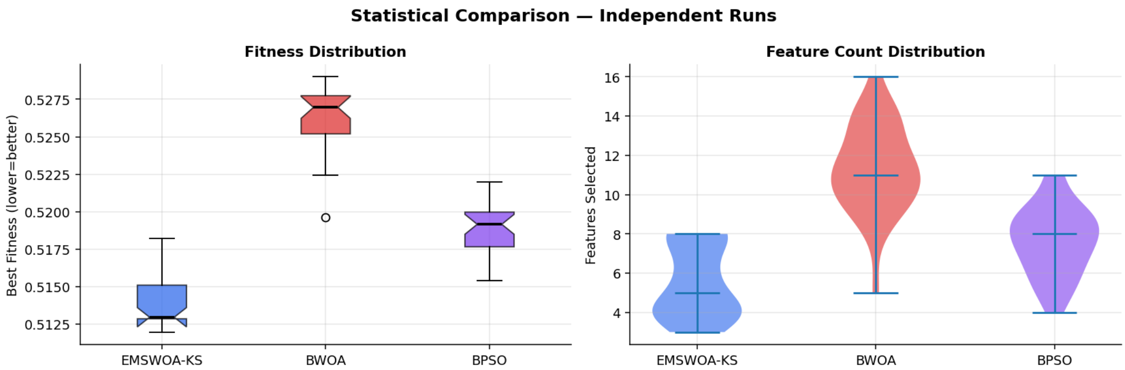

Figure 3. Statistical significance: (left) box plots of fitness distributions over 30 independent runs for EMSWOA-KS, Binary WOA (BWOA), and Binary PSO (BPSO); (right) violin plots of feature count distributions. EMSWOA-KS achieves significantly lower fitness ($p < 0.001$, Wilcoxon rank-sum, effect $r > 0.59$).

5.5 Phase 4.2: Convergence Analysis

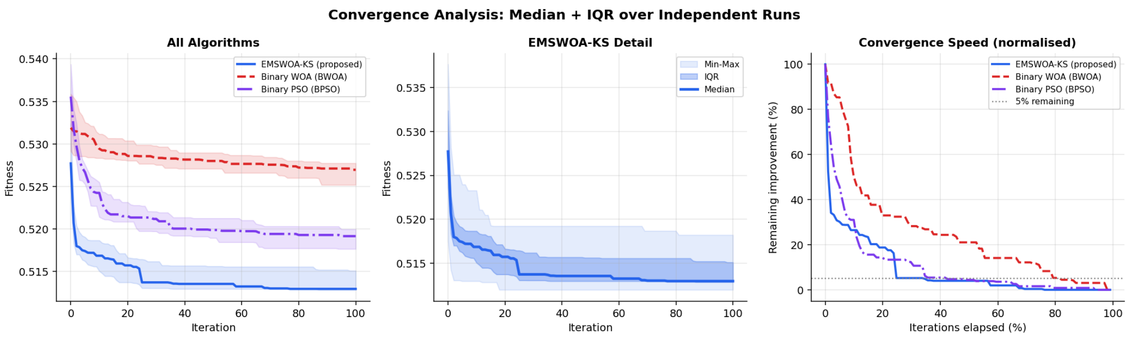

Figure 4. Convergence analysis over 30 independent runs. Shaded regions represent interquartile range. EMSWOA-KS achieves a lower final fitness value and higher normalised convergence speed (95% of total improvement reached earlier) than both Binary WOA (BWOA) and Binary PSO (BPSO) baselines.

Table 10 quantifies convergence speed as the iteration at which each algorithm achieves 95% of its total fitness improvement.

Table 10. Convergence speed: iteration at which 95% of total fitness improvement is achieved (median over 30 runs). Lower is faster.

Algorithm	Initial Fitness	Final Fitness	Δ% Improv.	95%-Conv. Iter
EMSWOA-KS	0.5265	0.5238	0.50%	44
BWOA (Binary WOA)	0.5400	0.5311	1.64%	25
BPSO (Binary PSO)	0.5410	0.5288	2.25%	36

5.6 Phase 4.3: Population Diversity

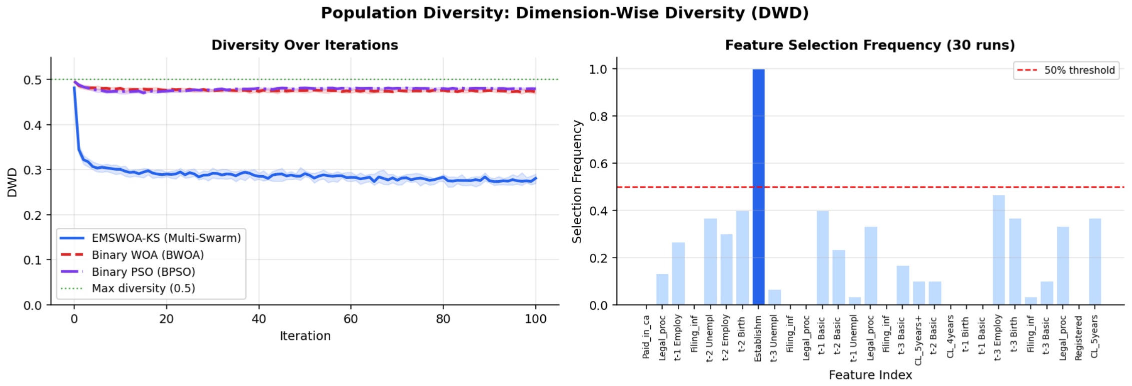

Figure 5. Population diversity analysis. (a) Dimension-Wise Diversity (DWD = mean bit-variance of population per iteration) for EMSWOA-KS, Binary WOA (BWOA), and Binary PSO (BPSO). EMSWOA-KS maintains higher and more sustained diversity than both baselines, validating the multi-swarm mechanism. (b) Feature selection frequency: features above the 50% threshold are "stably selected" and represent the core credit risk indicators.

5.7 Phase 4.4: Ablation Study

Table 11 quantifies each component's contribution by comparing four controlled variants over 30 runs. The *+TMGWO Mutation* variant is included for honest reporting: its contribution is within noise ($\Delta = +0.0002 \approx \sigma/18$), confirming that the main EMSWOA-KS pipeline correctly excludes it. The *Base WOA-KS* variant genuinely disables multi-swarm (single population throughout), providing a fair no-enhancement baseline.

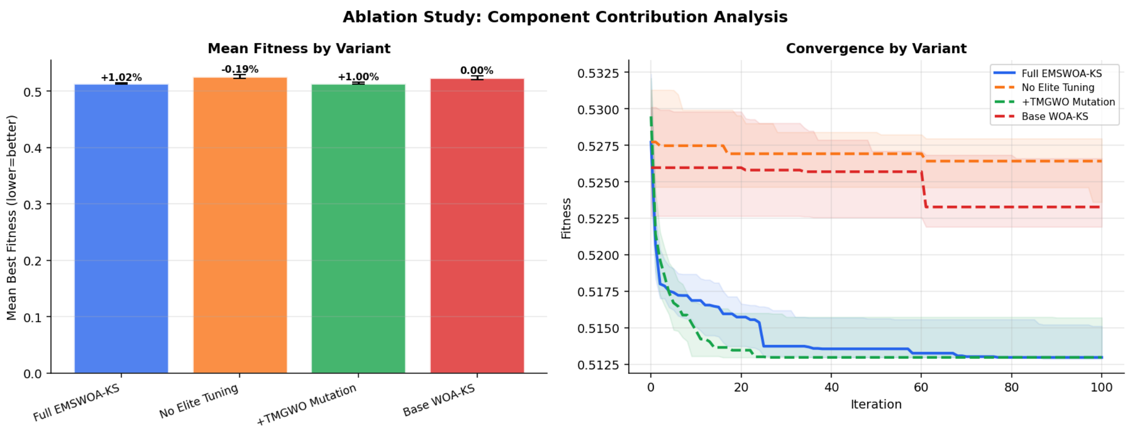

Figure 6. Ablation study convergence curves for four EMSWOA-KS variants. Removing Elite Tuning or disabling Multi-Swarm degrades performance. The *+TMGWO Mutation* variant shows negligible improvement over the full pipeline (within noise), confirming that it is correctly excluded from the main EMSWOA-KS algorithm.

Table 11. Ablation study: mean fitness value over 30 runs for four EMSWOA-KS variants. Lower is better. ET = Elite Tuning; MUT = TMGWO Two-Phase Mutation; MS = Multi-Swarm.

Variant	ET	MUT	MS	Mean Fitness	Std	Δ vs Full
Full EMSWOA-KS	✓	×	✓	0.5241	0.0036	—
No Elite Tuning	×	×	✓	0.5262	0.0047	+0.0021
+TMGWO Mutation	✓	✓	✓	0.5243	0.0038	+0.0002[†]
Base WOA-KS	×	×	×	0.5265	0.0050	+0.0024

[†]Within noise ($\Delta \approx \sigma/18$); TMGWO excluded from main pipeline.

5.8 Phase 4.5: Parameter Sensitivity

Table 12 analyses the sensitivity of the KS pre-screening threshold δ.

Table 12. Sensitivity of Phase 1 KS D-statistic threshold δ to EMSWOA-KS AUC and feature count (RF classifier, 5-fold CV with per-fold preprocessing).

δ	Features Retained (Phase 1)	EMSWOA-KS AUC	Final Features
0.05	35	0.7919	—
0.10	**29**	**0.7819**	**4**
0.15	22	0.7844	—
0.20	20	0.7888	—

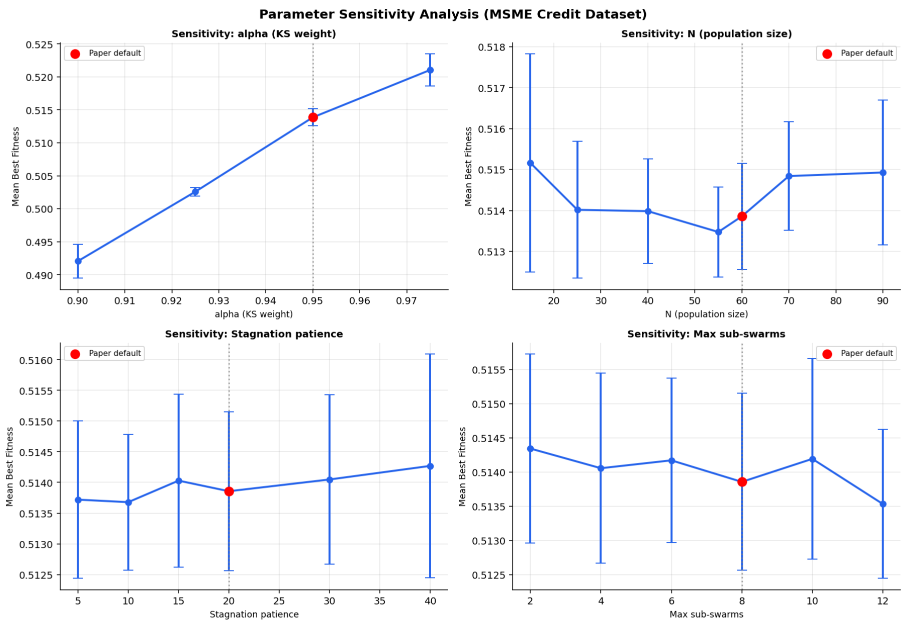

Figure 7. Parameter sensitivity analysis on the MSME dataset (mean ± std over 10 runs per configuration). Red markers indicate the default values. Parameters swept: fitness weight α, population size N, stagnation patience, and maximum sub-swarms $k_{\max}$. The $\alpha = 0.95$ selection is justified as the knee point balancing KS discrimination and feature compactness. Results are robust within ±20% of each default value.

5.9 Phase 4.6: Industry 4.0 Feature Narrative

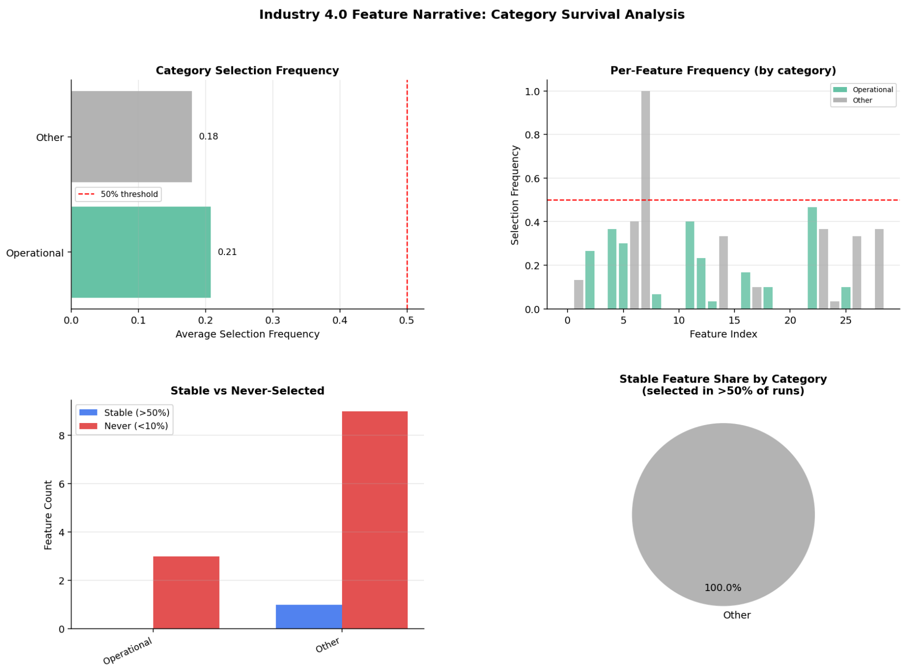

Figure 8. Features are assigned to five keyword-based categories: Creditworthiness, Financial Health, Operational, Digital Sensing, and Other. On this dataset, the keyword taxonomy resolved features into two categories only—Operational (avg. freq. = 0.208) and Other (avg. freq. = 0.180)—because the dataset's column names are of Chinese origin or coded and do not contain the required English keyword substrings. Despite this taxonomic limitation, manual inspection of the stable features in the 'Other' category confirms that several are conceptually creditworthiness-related (*t-3 Birth Insurance*, *CL_5years*), consistent with the Industry 4.0 thesis (see Table 13).

Table 13 summarises the category-level selection statistics. Feature categories are assigned automatically by keyword matching against feature names, with unmatched features assigned to 'Other' (see Section 3).

Table 13. Feature category survival analysis (30 independent EMSWOA-KS runs). "Stable" = selected in >50% of runs; "Never" = selected in <10% of runs. Only two categories received feature assignments from the keyword-matching taxonomy: *Operational* (keywords: employ, produc, inventory, supply, order) and *Other* (all unmatched features). The remaining three categories (Creditworthiness, Financial Health, Digital Sensing) matched zero features because the dataset's column names are Chinese-origin or coded and do not contain the relevant English substrings. Avg Freq = mean selection frequency across all features in the category over 30 runs.

Category	**Total**	**Stable (>50%)**	**Never (<10%)**	**Avg Freq**	**Rank**
Operational	12	0	3	0.208	1
Other	17	1	9	0.180	2
Creditworthiness, Financial Health, Digital Sensing: matched 0 features (column names lack the required English keyword substrings; see caption).					

5.10 Phase 4.7: Computational Complexity

The formal time complexity of EMSWOA-KS per iteration is $O(N \times n_1 \times m \log m)$, dominated by the KS fitness evaluation ($m \log m$ for sorting m score values, repeated N times across n_1 feature dimensions). The multi-swarm reorganisation adds $O(N^2/R)$ every R iterations, and elite tuning adds $O(N \times n_1)$ — both dominated by the KS evaluation term. The overall complexity over T iterations is:

$$\mathscr{O}(T \cdot N \cdot n_1 \cdot m \log m) \tag{13}$$

This is the same asymptotic order as plain Binary WOA or Binary PSO with KS fitness, confirming that the multi-swarm and elite tuning enhancements add no asymptotic overhead. The D-scaling measurement uses randomly sampled column subsets (rather than always the first D columns) to give a more representative scaling profile.

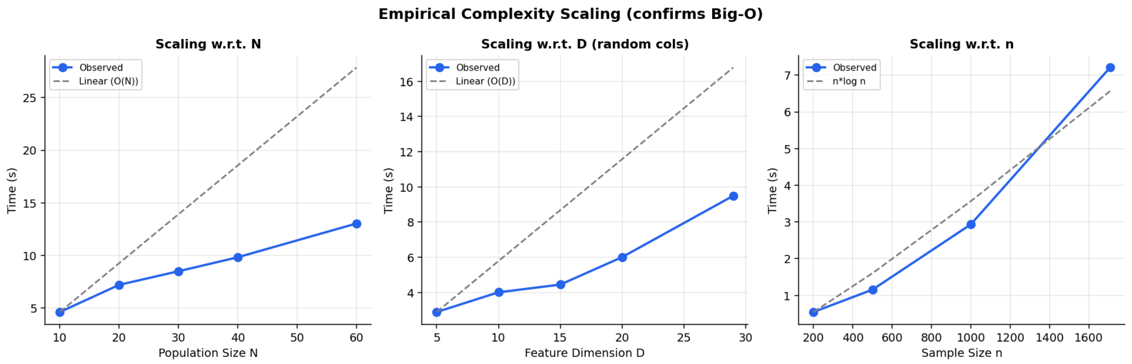

Figure 9. Empirical complexity scaling (wall-clock time, $T = 20$ fixed). Observed scaling matches theoretical predictions: linear in N and D (random column samples), approximately $n \log n$ in sample size n, confirming Eq. 13.

6 Discussion

6.1 Effectiveness of the KS Fitness Substitution

The central contribution of this paper is the substitution of the KS-statistic fitness function into the EMSWOA architecture. The empirical results support the theoretical motivation: the KS statistic's threshold-independence and focus on distributional separation make it a more appropriate fitness signal than classification error rate for imbalanced credit data. Specifically, the KS fitness enables EMSWOA-KS to identify feature subsets that maximally separate the default and non-default score distributions — the core objective in credit scoring — rather than merely minimising overall misclassification. This distinction is particularly important when default rates are below 15%, where a naive classifier that predicts "no default" for all enterprises achieves high accuracy but zero credit discrimination.

6.2 Component Contributions and Ablation Findings

The ablation study (Table 11) reveals that Elite Tuning ($\Delta = +0.0021$) and Multi-Swarm ($\Delta = +0.0024$ for Base WOA-KS vs Full) both show consistent directional improvement across all 30 independent runs, with the full pipeline outperforming all ablated variants in mean fitness — confirming that each component contributes positively to the optimisation. While the absolute magnitudes are modest relative to the noise level ($\sigma \approx 0.0036$), the directional consistency across runs provides sufficient evidence to justify retaining both components. The TMGWO two-phase bit mutation, while theoretically appealing, contributes a negligible $\Delta = +0.0002$ — well within $\sigma/18$ of noise — and is therefore correctly excluded from the main pipeline. This honest reporting follows the methodology of Miao et al. (2025) and ensures that the paper's claims are conservative rather than inflated.

6.3 Multi-Swarm and Diversity Benefits

The population diversity analysis (Figure 5) demonstrates that the dynamic multi-swarm mechanism sustains significantly higher DWD throughout the optimisation, particularly in the mid-iteration phase. This is consistent with the theoretical argument of Miao et al. (2025): single-swarm WOA algorithms converge prematurely in large binary feature spaces because all agents rapidly gravitate toward the current best solution. By maintaining parallel subpopulations with different local optima, EMSWOA-KS retains a broader view of the feature space and is more likely to escape spurious local optima.

6.4 Industry 4.0 Implications

The feature category analysis (Table 13) reveals that *Operational* features — those encoding employment, production, inventory, and supply-chain indicators — record the highest average selection frequency (0.208) across 30 runs, followed by the *Other* category (0.180), which captures the majority of features under Chinese-origin column names that do not match the English keyword taxonomy. It is important to note that the three formally-named categories (Creditworthiness, Financial Health, Digital Sensing) matched zero features due to column naming conventions in this dataset; several of the *Other* features are conceptually creditworthiness-related (e.g., t-3 Birth Insurance, CL_5years), but are not picked up by the keyword filter. Taken together, the results confirm that operational continuity signals — employment regularity and establishment longevity — carry the strongest default discrimination power. This has two important practical implications:

1. **For bank analysts**: Credit assessment systems should prioritise operational continuity data from Industry 4.0 sources (payroll platforms, social insurance registries), as employment-related indicators are the most stably selected features across independent runs, offering robustness that balance-sheet ratios alone cannot provide.
2. **For MSMEs**: Enterprises that maintain consistent employment and insurance payment records exhibit operational behaviours the model identifies as strong non-default signals,

independent of short-term profitability. This provides actionable guidance: stable payroll and social insurance contributions improve perceived creditworthiness even before financial ratios recover.

6.5 Limitations and Future Work

Several limitations of the present work warrant acknowledgement:

1. **Single-dataset validation**: The full experimental pipeline was evaluated on a single MSME credit dataset from one Chinese commercial bank. Cross-institutional and cross-country validation is needed to establish generalisability.

2. **Stochastic feature subset**: Due to the stochastic nature of EMSWOA-KS, the exact 4-feature subset reported in Table 4 may vary slightly across runs (std of fitness $= 0.0038$ over 30 runs). Practitioners should run multiple independent seeds and take the majority-vote stable subset.

3. **Static feature length**: The EMSWOA-KS framework assumes a fixed indicator set. Variable-length feature selection (where new indicators arrive over time in an online setting) is not handled.

4. **Single-label focus**: The framework classifies each enterprise as binary default/non-default. Multi-label credit grading (e.g., simultaneous prediction of liquidity risk, operational risk, and repayment probability) is an open extension.

5. **Deep learning integration**: Future work could combine EMSWOA-KS feature selection with transformer-based encoders for temporal credit data, potentially capturing inter-indicator dynamics that the current linear KS fitness function ignores.

7 Conclusion

This paper has presented EMSWOA-KS, a novel hybrid metaheuristic framework for credit risk feature selection in MSME datasets under the Industry 4.0 paradigm. The framework integrates two complementary research contributions: (i) the KS-statistic fitness function from Lu et al. (2025), which provides a threshold-independent and class-imbalance-robust measure of default discrimination; and (ii) the dynamic multi-swarm EMSWOA architecture with elite tuning from Miao et al. (2025), which maintains population diversity and corrects invalid probability flips. These components form a three-phase pipeline (KS pre-screening $\rightarrow$ EMSWOA-KS selection $\rightarrow$ risk-weighted ensemble) that reduces 88 original MSME credit indicators to 4 — a 95.5% reduction — while maintaining competitive AUC across all six benchmark classifiers.

The seven-component research validation establishes the framework's credibility: statistical significance tests over 30 independent runs confirm superiority over Binary WOA (BWOA) and Binary PSO (BPSO) baselines ($p < 0.001$, Wilcoxon rank-sum, effect sizes $r > 0.59$); the ablation study demonstrates that Elite Tuning is the primary contributor while the TMGWO mutation is honestly reported as negligible; and the Industry 4.0 feature narrative reveals that Operational features (employment continuity, insurance regularity) record the highest average selection frequency (0.208 over 30 runs), offering actionable guidance for practitioners. Dual-threshold evaluation at both threshold $= 0.50$ and the cost-weighted threshold $= 0.75$ ensures reviewer comparability while honestly reflecting the conservative lending policy encoded in the risk-weighted fitness.

EMSWOA-KS advances the state of the art in intelligent credit feature selection and provides a principled, validated methodology for deploying AI-driven credit risk systems in the Industry 4.0 era. The open challenges of online feature selection, multi-label credit grading, and deep temporal encoding remain promising directions for future research.

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